Game Theory
Lecture 2: Strategic form games and NE[1]

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Outline

1. Strategic games
   - Notation
   - Nash equilibrium
     - NE in pure strategies
     - Why Nash equilibrium?
2. Mixed Strategies

An example I

		Player 2	
		C	D
Player 1	C	$2, 2$	$0, 3$
	D	$3, 0$	$1, 1$

Table: prisoners dilemma

- What do the numbers in the game table actually mean?
- What if the other player plays C and D with 50% probability? How to evaluate that?
- Dominant strategy?
- Nash equilibrium?

Notation I

- finite set N of players; usually we call the players "player 1",“player 2”, “player 3” or simply “P1”, “P2” etc.
- set of actions A_i from which each player i has to choose; if all A_i are finite, we say the game is finite
- we call $a = (a_i)_{i \in N}$ with $a_i \in A_i$ an *action profile* (and sometimes an *outcome*), i.e. each player takes one of his actions;
  sometimes it is convenient to denote an action profile by (a_i, a_{-i})
- $A = \times_{i \in N} A_i$ the set of all action profiles/outcomes
- each player i has a preference $\succeq_i$ over the outcomes in A; we will normally assume that this preference relation can be represented by v.N-M. expected utility function $u_i : A \to \mathbb{R}$;
  we refer to values of u_i as *payoffs*

Notation II

- sometimes the utility function will also depend on a random variable $\omega \in \Omega$, then $u_i : A \times \Omega \to \mathbb{R}$

- a strategic game can be denoted by $\langle N, (A_i), (u_i) \rangle$

A convenient way to write 2-player strategic games is a table. Normally, "player 1" (P1) is the row player and "player 2" (P2) is the column player.

	L	R
T	w_1, w_2	x_1, x_2
B	y_1, y_2	z_1, z_2

Table: 2-player game written in table

- What is w_1 in our game notation?
- What are the action sets?

Notation III

- Give an example of an action profile.

Notation IV

Example (Cournot competition)

- 3 firms
- each firm chooses a quantity q_i, $i = 1, 2, 3$, it wants to put on the market
- the market price will be $p = max\{1 - q_1 - q_2 - q_3, 0\}$
- assume each firm's costs are zero
- each firm maximizes its expected profits

Questions:

- What is the set of players N?
- What is the action set A_i?
- Give an example of an action profile.
- What are the player's utility functions u_i?

Interpretation of a game

Interpretation of a strategic game:

1. one shot game
   - one time event
   - each player knows game
   - rationality is common knowledge
   - actions are chosen simultaneously and independently
   - a player can base his expectation of other player's play only on primitives of the game
2. repeated play without strategic link
   - game is played repeatedly but with different opponents
   - no intertemporal strategic link between games
   - players can have expectations how rational players play based on past plays

Nash equilibrium I

Steady state concept:

- each player has correct expectation about other players' behavior
- each player maximizes his utility given his expectation of other players' behavior (rational)

Definition (Nash equilibrium)

A Nash equilibrium of a strategic game $\langle N, (A_i), (u_i) \rangle$ is an action profile $a^* \in A$ such that for every player $i \in N$

$$u_i(a^*_{-i}, a^*_i) \geq u_i(a^*_{-i}, a_i) \qquad \text{for all } a_i \in A_i.$$

Nash equilibrium II

Definition (best response)

The correspondence B_i defined by

$$B_i(a_{-i}) = \{a_i \in A_i : u_i(a_{-i}, a_i) \geq u_i(a_{-i}, a_i') \text{ for all } a_i' \in A_i\}$$

is player i's *best response*.

Example

	L	M	R
U	0,1	2,1	1,0
D	1,1	2,2	0,0

- What is P2's best response to U, i.e. $B_2(U)$?
- What is $B_2(D)$?

Nash equilibrium III

- Note: Sometimes it will be convenient to write B_i as a function of a instead of a_{-i}.
- A Nash equilibrium is an action profile a^* for which $a_i^* \in B_i(a_{-i}^*)$ for all $i \in N$.
- Define $B(a) = B_1(a) \times B_2(a) \times \cdots \times B_n(a)$. A NE is now an a^* such that $a^* \in B(a^*)$.
- method to compute NE: Calculate all best responses and construct the set valued function $B(a)$; search for fixed points of $B(a)$.

Nash equilibrium IV

Exercise

Determine best responses and Nash equilibria of the following game:

	Go 2	Stop 2
Go 1	-1,-1	4,1
Stop 1	1,4	3,3

Table: Chicken

Why is Nash equilibrium reasonable?

- self-enforcing recommendation,
- steady state interpretation of game
- pre-play negotiations: if players make non-binding agreements before play, which agreements will actually be honored? NE!
- (evolutionary stable strategy)

Mixed Strategies I

Example (mixed strategy equilibrium: matching pennies)

	H	T
H	1,0	0,1
T	0,1	1,0

Table: Matching Penies

- game has no pure strategy equilibrium
- how would you play this game?

Mixed Strategies II

- sometimes it is best to randomize
- extend out strategy space: a strategy of player i is a probability distribution over the elements of A_i
- denote by $\Delta(A_i)$ the set of probability distributions over A_i
- a member of $\Delta(A_i)$ is a mixed strategy of player i
- take $\delta \in \Delta(A_i)$ and let A_i be finite
  - $\delta(a_i)$ is the probability that δ assigns to a_i
  - set of $a_i \in A_i$ such that $\delta(a_i) > 0$ is the *support* of δ
- $a_i \in A_i$ is a *pure strategy*, i.e. a mixed strategy that puts all probability weight on a_i
- for finite games: under a profile $(\alpha_j)_{j \in N}$ of mixed strategies the probability of the action profile $a = (a_j)_{j \in N}$ is $\Pi_{j \in N} \alpha_j(a_j)$

Mixed Strategies III

- player i's preferences are represented by the expected utility function $U_i : \times_{j \in N} \Delta(A_j) \to \mathbb{R}$ which assigns to $(\alpha_j)_{j \in N}$ the value

$$U_i(\alpha) = \sum_{a \in A} \left(\Pi_{j \in N} \alpha_j(a_j)\right) u_i(a).$$

- give an example for a mixed strategy of P1 in matching pennies
- give an example for a pure strategy of P1 in matching pennies
- what is $\Delta(A_1)$ in the matching pennies example?

Mixed Strategies IV

Example

- 3 player game
- give an example of a mixed strategy of P1
- what is P1's expected payoff with this mixed strategy if P2 plays L and P3 plays B?
- what is P1's expected payoff with this mixed strategy if
  - P2 plays L with probability 2/3 and R with probability 1/3
  - P3 plays A and B with probability 1/2?

	L	R
H	1,0,1	0,1,0
T	0,1,1	1,0,0

A

	L	R
H	1,0,0	0,1,1
T	0,1,0	1,0,1

B

Mixed Strategies V

Definition

The *mixed extension* of the strategic game $\langle N, (A_i), (u_i) \rangle$ is the strategic game $\langle N, (\Delta(A_i)), (U_i) \rangle$.

Definition

A *mixed strategy Nash equilibrium* of a strategic game G is a Nash equilibrium in the mixed extension of G.

- any suggestions for a mixed strategy equilibrium in matching pennies?

Mixed Strategies VI

Lemma

$\alpha^* \in \times_{i \in N} \Delta(A_i)$ is a mixed strategy Nash equilibrium of $\langle N, (A_i), (u_i) \rangle$ if and only if for every player i, every pure strategy in the support of α_i^* is a best response to α^*_{-i}. Hence, any action played with positive probability in a mixed strategy NE yields the same payoff.

Proof.

Mixed Strategies VII

Example (mixed strategy equilibrium: chicken)

	Go 2	Stop 2
Go 1	-1,-1	4,1
Stop 1	1,4	3,3

Table: Chicken

This game has two NE in pure and one in completely mixed strategies. Find all three!

Interpretation of mixed strategies I

- deliberate randomization (e.g. Poker)
  problem: motivation for randomization seems to be out-guessing/avoiding other player which seems to be deliberate and not random (psychological)
- stochastic steady state
  frequencies that are necessary to keep system steady
- non-modelled aspects. The action of a player depends on things that are not modelled
  Problem: model incomplete?! or deliberate behavior depends on factors that have nothing to do with payoff

Interpretation of mixed strategies II

- Players play pure strategies but opponents do not know which; mixed strategy is belief of other players not actual randomization $\Rightarrow$ equilibrium is a steady state of players' beliefs about others' actions
  Problem: all opponenets have same belief! predictive power of equilibrium appears smaller
- nice mathematical tool that guarantees equilibrium existence in finite games
  Problem: some interpretation and connection to strategic behavior would be nice
- small uncertainty about other players payoff (purification)

Review Questions

- What is a strategic form game? Which elements constitute a strategic form game?
- What is the difference between an action and an action profile?
- What are the two (main) interpretations of a strategic form game?
- Write down the formal definition of a Nash equilbrium and of a best response!
- Why is Nash equilibrium an interesting concept?
- What are mixed strategies and how can we denote them? What is the mixed extension of a game?
- How can we interpret mixed strategies?

reading: OR 2.1-2.3, 3.1-3.2 or MSZ 4.1,4.3,4.4,4.8,4.9, 5.1-5.2

Exercises I

1. This exercise is about a simple the Bertrand game: There are two firms trying to sell the same good to one consumer. The two firms offer a price and the consumer will buy the good at the lower of the two prices if the price is below his valuation 1. The firms have zero costs. Write this formally as a game, i.e. define a set of players, set of actions, utility functions. Find a Nash equilibrium of this game and check that it satisfies the conditions in the definition of Nash equilibrium.
2. Assume now that in the Bertrand game above prices have to be quoted in whole cents. Suppose that firm 1 is playing a mixed strategy putting weight $\alpha > 0$ on p_1 and weight $1 - \alpha > 0$ on p_2. Assume $0.01 < p_1 < p_2 < 1$. What is firm 2's best response?

Exercises II

3. Determine a mixed strategy equilibrium (and two pure strategy equilibria) in the following game:

	L	M	R
U	5,7	2,1	8 ,5
D	0,1	5,3	1 ,0